



**Optimize enterprise AI
storage for high-performance
and robust security with F5**



95%
**of generative AI
pilots at companies
are failing***

Source: [MIT Project NADA](#), [Fortune](#)



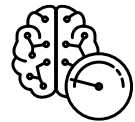
Poor data quality

Inaccurate and inconsistent data compromise AI models.



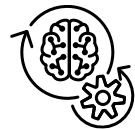
Lack of data governance

Lack of governance for data collection, storage, and usage creates new risks.



Inefficient data infrastructure

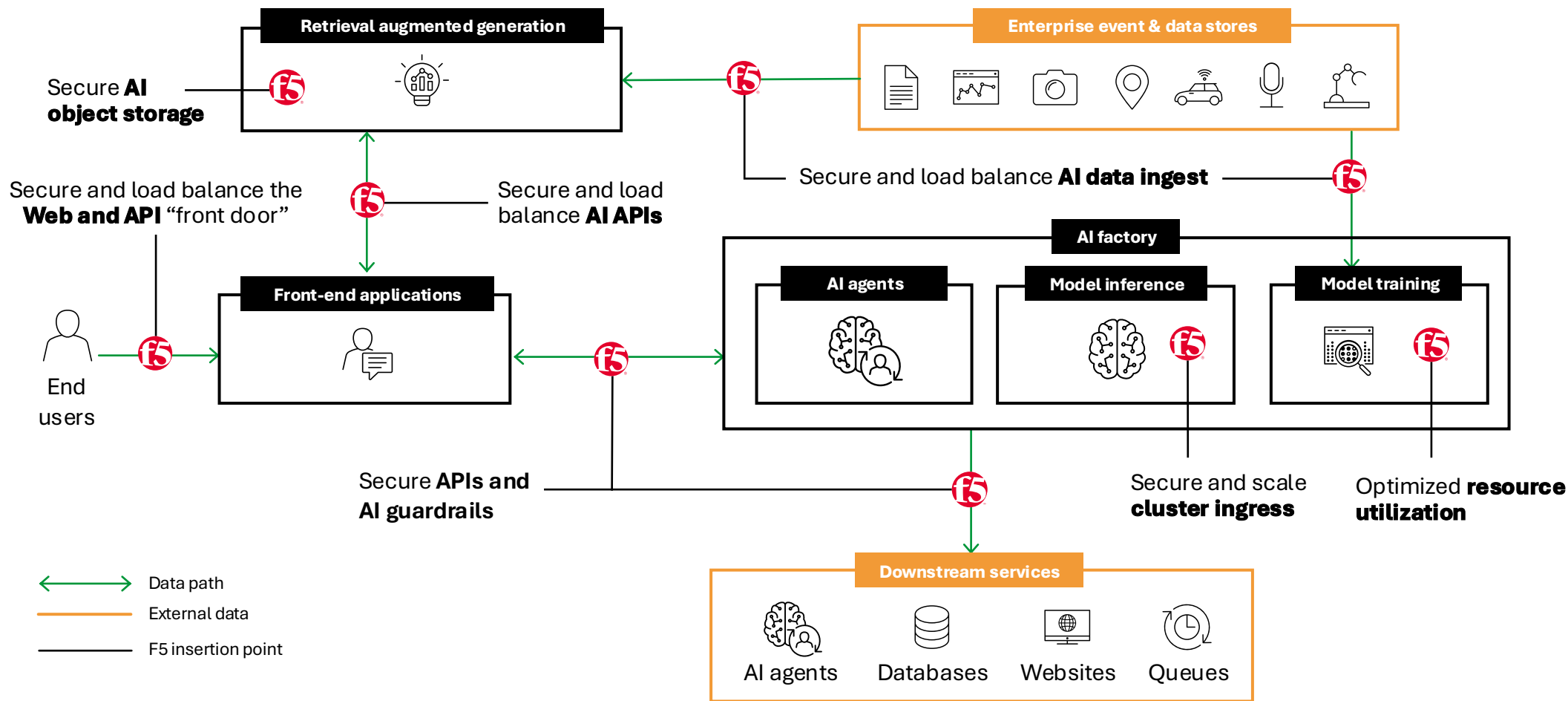
Insufficient data delivery speed at scale limits AI model potential.



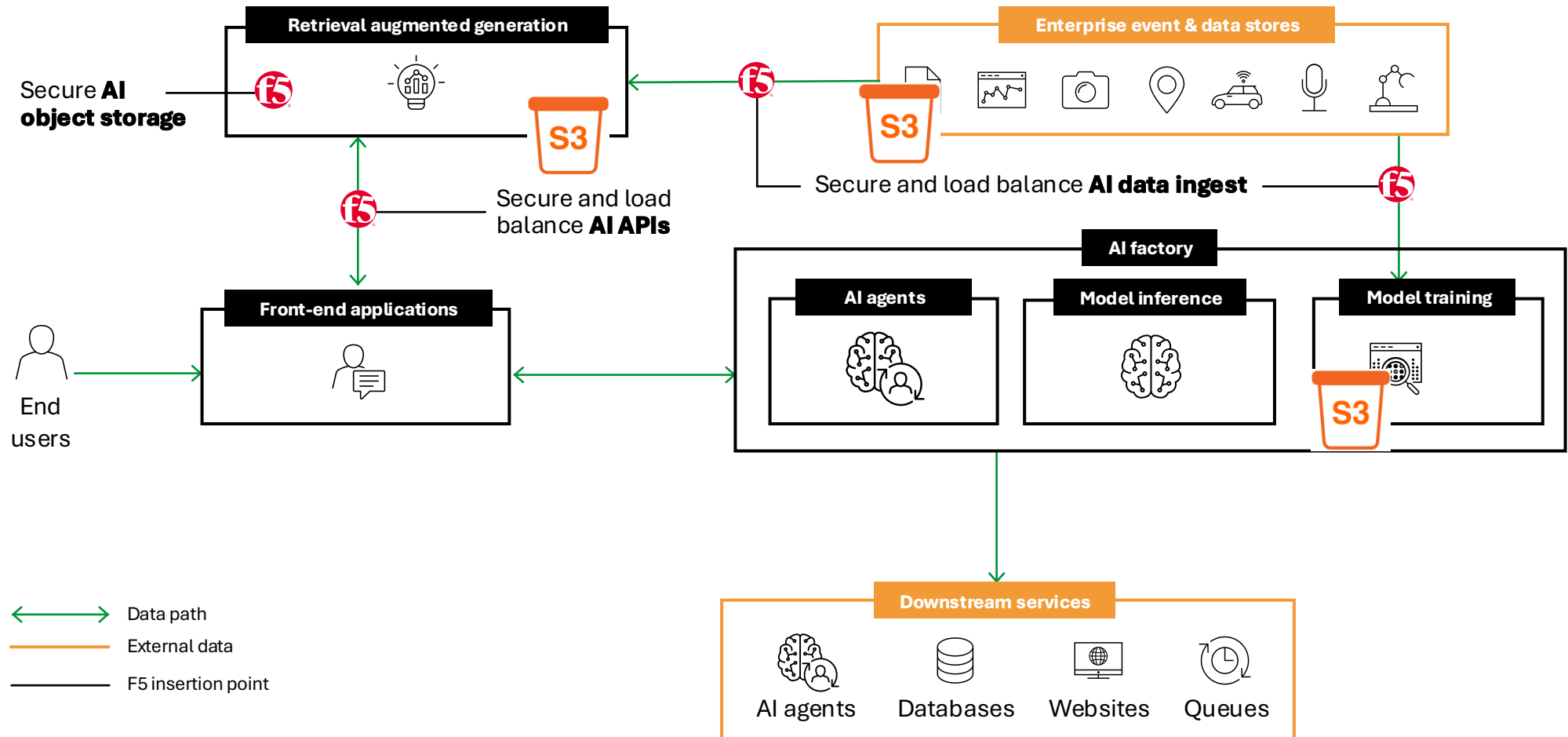
AI tech and operations are misaligned

Seamless integration into existing workflows increase the success of AI initiatives.

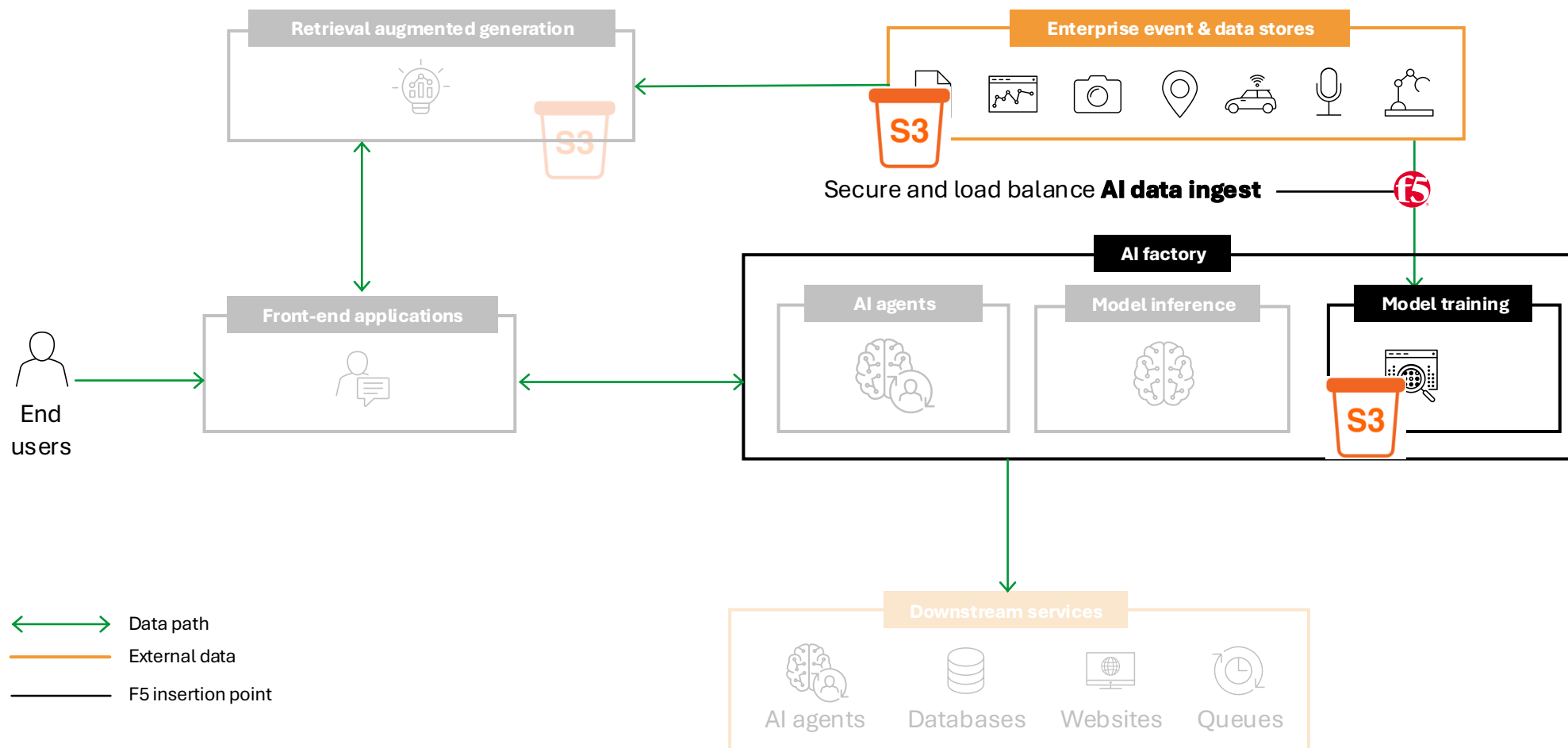
F5 enables enterprise scale delivery and security for AI application and model foundry workflows



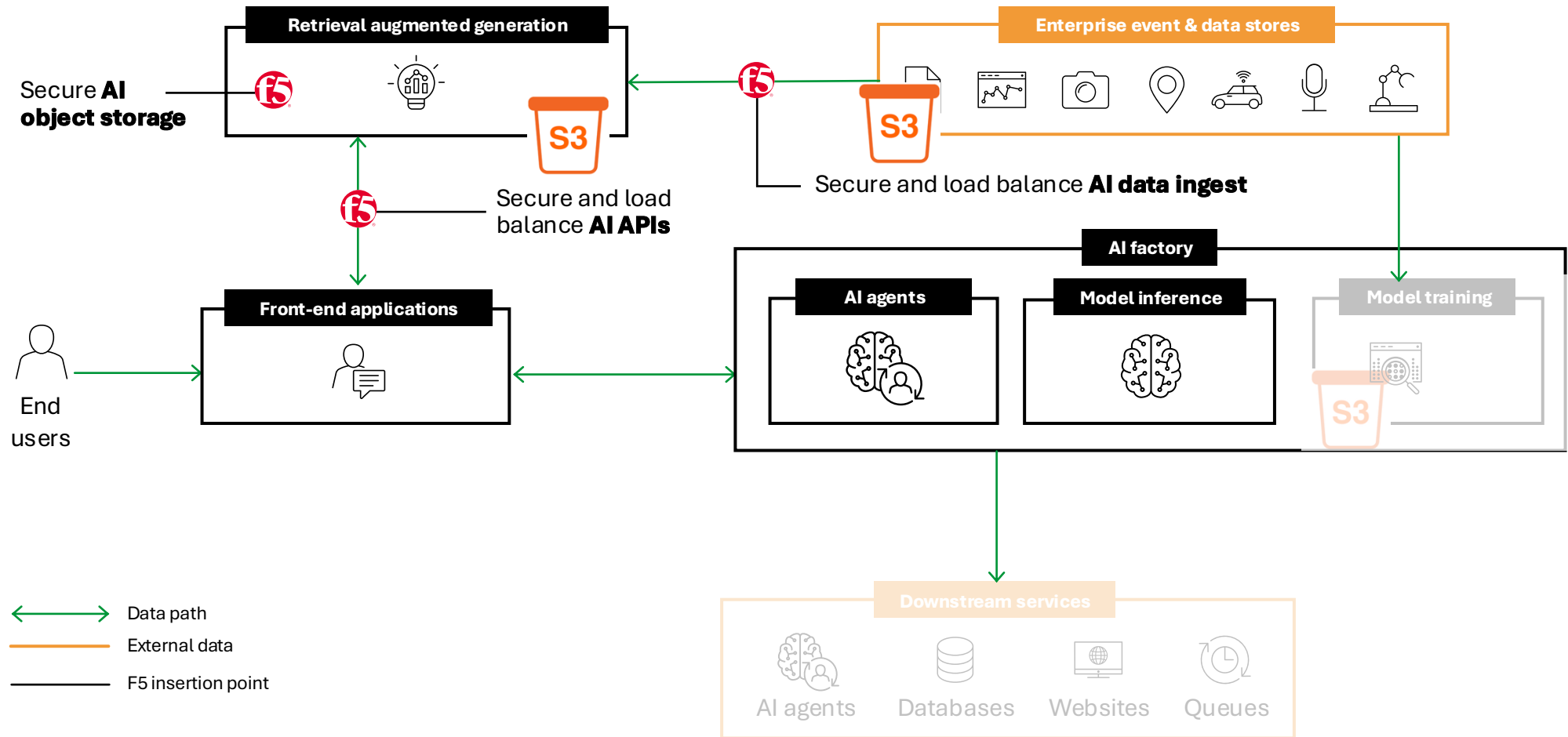
F5 + NetApp ensures fast, secure, and reliable AI data delivery to fuel AI pipelines at scale



AI model training is delivered securely at scale with F5 + NetApp



AI model inference is delivered securely at scale with F5 & NetApp



F5 BIG-IP best-in class traffic management & security for S3

Speed

Smart load balancing & security

Directs traffic to the optimal storage for performance, security, and availability

Seamless data flow

BIG-IP LTM ensures efficient, secure routing from external sources to local storage

Optimized S3 routing

BIG-IP DNS directs client connections to highly available storage nodes for smooth data ingestion

Scale

High-throughput traffic management

Optimize TCP and HTTPS flows for seamless object storage access

Accelerated packet processing

Leverage best in class FPGA and CPUs to accelerate packet processing for traffic management

Crypto offload for speed

BIG-IP LTM offloads encryption to best-in-class hardware on rSeries and VELOS, boosting performance

Security

Robust DDoS protection

BIG-IP AFM defends against volumetric and targeted attacks

Secure traffic management

BIG-IP LTM ensures efficient, secure routing from external sources to local storage

End-to-end data protection

Safeguards AI workloads with policy-driven security and threat mitigation

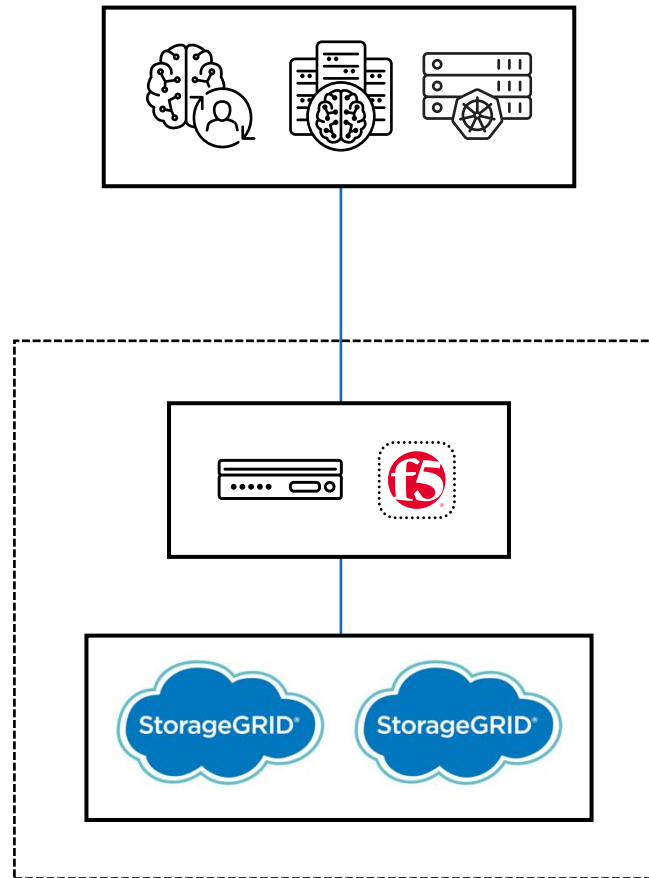
F5 for S3 resilience

Load balancing

Distribute load across multiple nodes or buckets to increase availability and scale based on policy

Health monitoring

Monitor health, connections and performance to check the status and capacity of nodes and redirect connections



Traffic programmability

Make traffic steering decisions based on information in the dataflow

Connection management

Mirror connection and persistence information to prevent interruption in service in the event of a failure

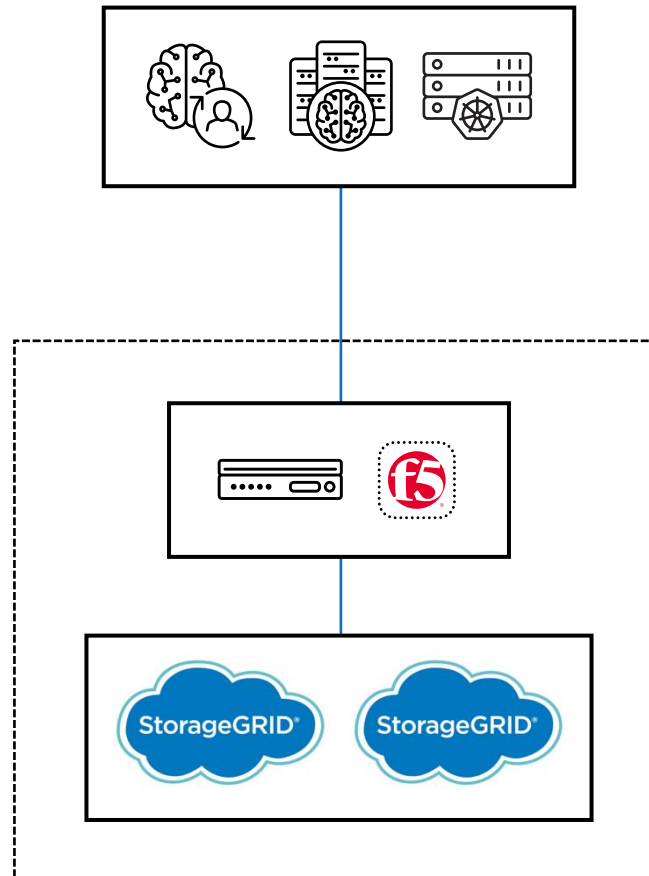
F5 for S3 performance optimization

Hardware acceleration

Purpose built FPGAs designed to accelerate traffic, increase throughput and response time

Bandwidth management

Distribute load based concurrent connections, bandwidth requirements or apply per-request policies



Connection optimization

Reduce backend connections using “OneConnect” Technology to increase node capacity

TCP optimization

Enable state-of-the-art optimization to dramatically improve network performance for different connection types

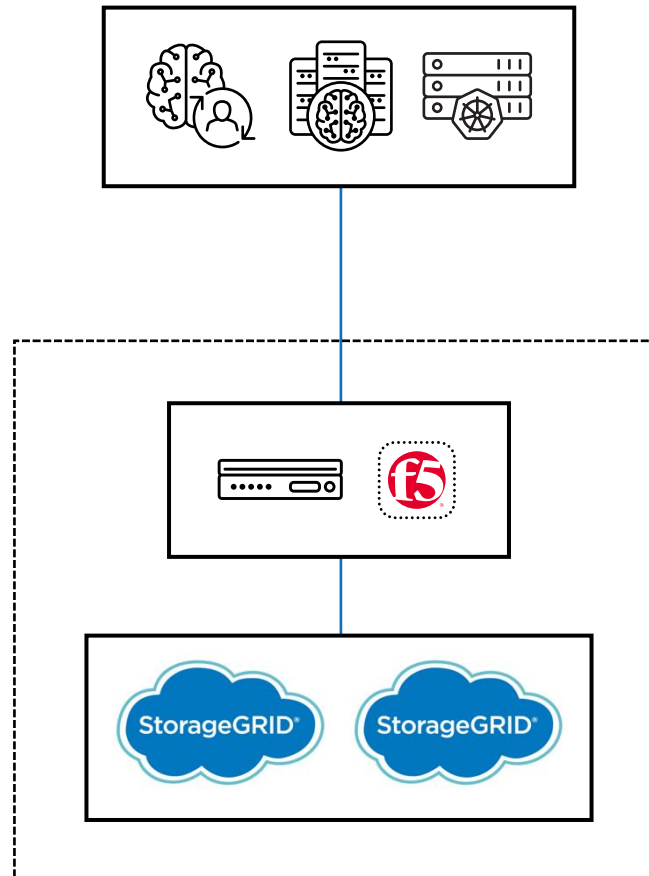
F5 for S3 security

Encryption offload

Purpose hardware to accelerate data decryption, offloading backend infrastructure and enforcing encryption standards

DDOS mitigation

Service provider scale DDOS mitigation in every appliance, ensuring maximum availability



Firewall and IPS

Integrated network firewall with IPS functionality including HTTP protocol validation and optional API security

App and API security

Advanced content inspection, API security and sensitive data masking

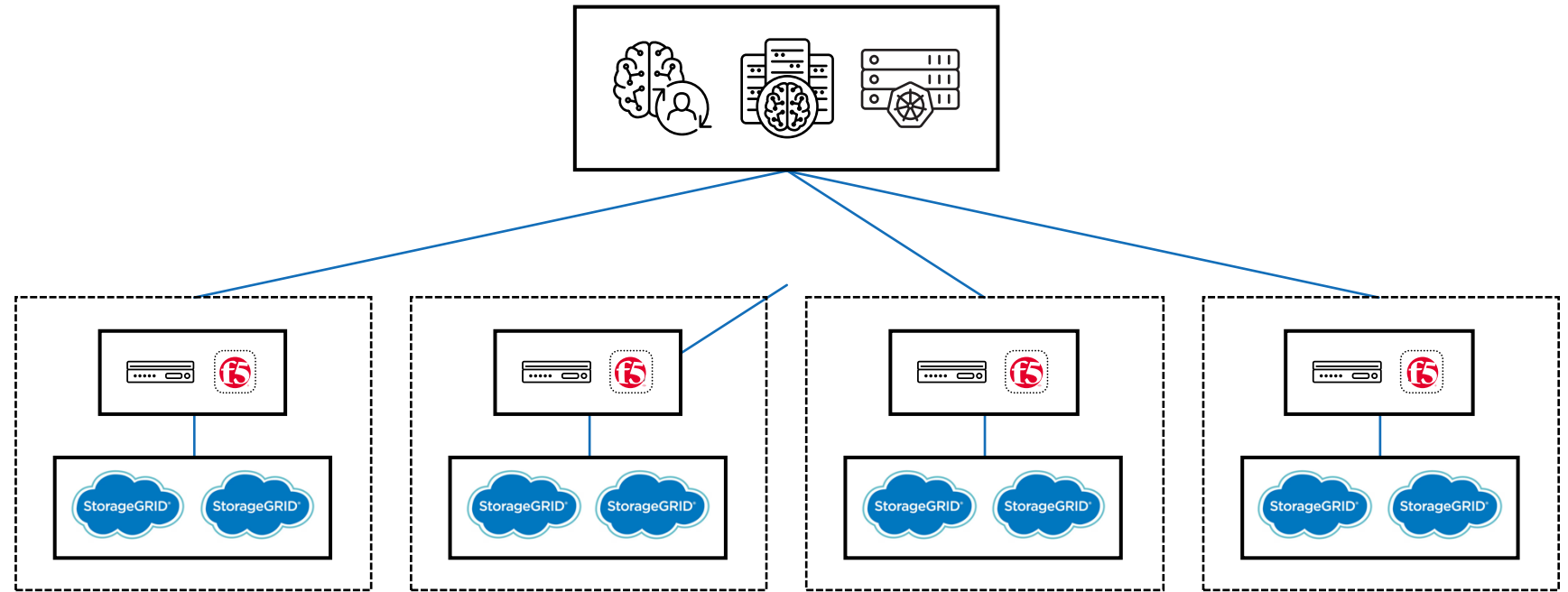
F5 for S3 global availability

Global load balancing

Ensure global availability across all your data centers and public clouds with automatic failover

Traffic steering

Direct connections to the closest or best performing location

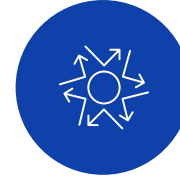


Why quantum-resistant S3 matters?



Strategic risk of inaction today

- **Data exposure risk:** Encrypted data in transit today may be decrypted by quantum technologies in the future, putting long-term assets at risk.
- **Business continuity threat:** Sensitive records (e.g., healthcare, financial, legal) stored in S3 could be compromised in the future, disrupting operations and trust.
- **Intellectual property vulnerability:** AI training data and proprietary insights stored in S3 may be exploited if decrypted, eroding competitive advantage.
- **Regulatory & legal exposure:** Emerging mandates (e.g., [Quantum Cybersecurity Act](#)) may require quantum-safe encryption—noncompliance could lead to penalties or litigation.



Mitigation strategy

- **F5 BIG-IP** enforces hybrid post-quantum cryptography (PQC) using NIST-approved ciphers (e.g., Kyber768).
- Offload PQC compute to **F5 hardware**, preserving NetApp storage performance.
- Use **F5 secure multicloud networking + NetApp SnapMirror** for encrypted, policy-driven data mobility.
- Enforce **TLS 1.3 with PQC** at the edge via F5, ensuring regulatory alignment for long-lived S3 data.

F5 Distributed Cloud Services provides modern day highways, connecting data locations with secure access

- Provides data storage mobility across zones and regions, regardless of where it's stored
- Quickly and securely connect private corporate data and enable restricted access
- Increases networking simplification and empowers data location silos



AI multicloud networking

Enable seamless and secure data and model connectivity across distributed AI environments.

